

- 2.1** Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4				
2	2				
1	3				
-1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum (x_i - \bar{x}) =$	$\sum (x_i - \bar{x})^2 =$	$\sum (y_i - \bar{y}) =$	$\sum (x_i - \bar{x})(y_i - \bar{y}) =$

- Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?
- Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.
- Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.
- Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.
Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4				
2	2				
1	3				
-1	1				
0	0				
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i \hat{e}_i =$

- On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.
- On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.
- Show that for these numerical values $\bar{y} = b_1 + b_2 \bar{x}$.
- Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i / N$.
- Compute $\hat{\sigma}^2$.
- Compute $\widehat{\text{var}}(b_2 | \mathbf{x})$ and $\text{se}(b_2)$.

a.

	x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(y - \bar{y})^2$	$(x - \bar{x})(y - \bar{y})$
	3	4	2	4	2	4	4
	2	2	1	1	0	0	0
	1	3	0	0	1	1	0
	-1	1	-2	4	-1	1	2
	0	0	-1	1	-2	4	2
sum	5	10	0	10	0	10	8
	$\bar{x} =$	1					
	$\bar{y} =$	2					

b.

$$b_2 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = 0.8$$

代表 x 每提升一個單位，平均 y 增加 0.8

$$b_1 = \bar{y} - b_2 \bar{x} = 1.2$$

代表當 x 為 0 時，y 平均為 1.2

c.

$$\sum x_i^2 = 15$$

$$\sum x_i y_i = 18$$

To show the equivalence

$$\sum (x_i - \bar{x})^2 = 10$$

$$\sum x_i^2 - N\bar{x}^2 = 15 - 5 * 1 = 10$$

And to show the equivalence

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$$

$$\sum x_i y_i - N\bar{x}\bar{y} = 18 - 5 * 1 * 2 = 8$$

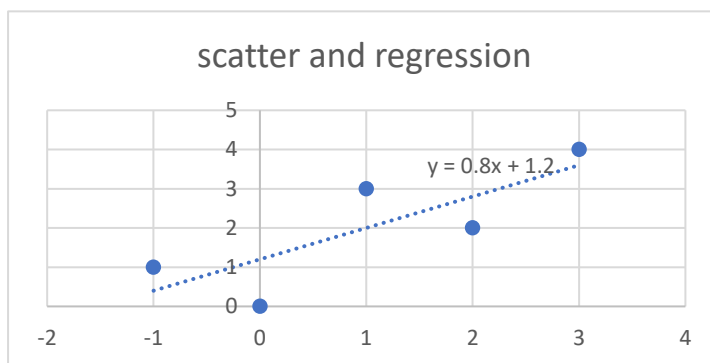
d.

$$\hat{y}_i = b_1 + b_2 x_i = 1.2 + 0.8 x_i$$

$$\hat{e}_i = y_i - \hat{y}_i$$

	x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
	3	4	3.6	0.4	0.16	1.2
	2	2	2.8	-0.8	0.64	-1.6
	1	3	2	1	1	1
	-1	1	0.4	0.6	0.36	-0.6
	0	0	1.2	-1.2	1.44	0
sum	5	10	10	0	3.6	-1.3E-15
	$s_y^2 =$	2.5				
	$s_x^2 =$	2.5				
	$s_{xy} =$	2				
	$r_{xy} =$	0.32				
	$CV_x =$	158.1139				
	$x_{50\%th}:$	1				

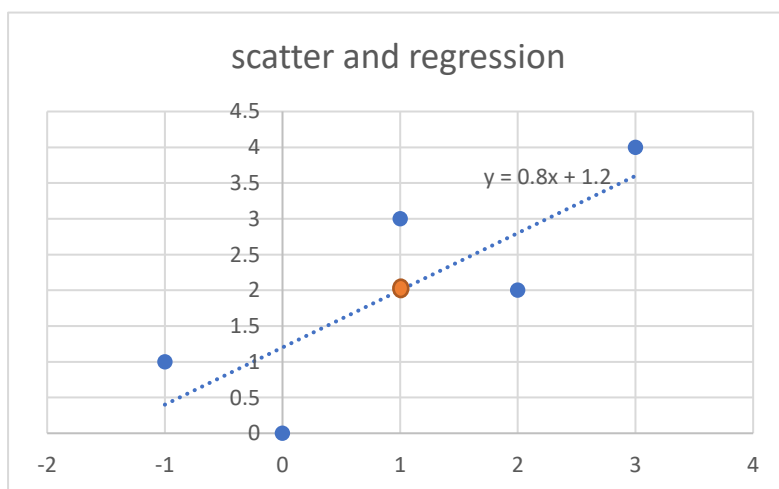
e.



f.

$$(\bar{x}, \bar{y}) = (1, 2)$$

位於回歸線之上



g. show that $\bar{y} = b_1 + b_2 \bar{x}$

$$\bar{y} = 2$$

$$b_1 + b_2\bar{x} = 0.8 + 1.2 * 1 = 2$$

h.

$$\bar{\hat{y}} = \frac{1}{5} \sum \hat{y} = \frac{3.6 + 2.8 + 2 + 0.4 + 1.2}{5} = 2 = \bar{y}$$

i.

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{5 - 2} = 1.2$$

j.

$$\widehat{var}(b_2|x) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$$

$$se(b_2) = \sqrt{\widehat{var}(b_2|x)} = 0.34641$$

2.22 A longitudinal experiment was conducted in Tennessee beginning in 1985 and ending in 1989. A single cohort of students was followed from kindergarten through third grade. In the experiment children were randomly assigned within schools into three types of classes: small classes with 13–17 students, regular-sized classes with 22–25 students, and regular-sized classes with a full-time teacher aide to assist the teacher. Student scores on achievement tests were recorded as well as some information about the students, teachers, and schools. Data for the kindergarten classes are contained in the data file *star5_small*. [Note: The data file *star5* contains more observations and variables.]

- a. Using children who are in either a regular-sized class or a small class, estimate the regression model explaining students' combined aptitude scores as a function of class size, $TOTALSCORE_i = \beta_1 + \beta_2 SMALL_i + e_i$. Interpret the estimates. Based on this regression result, what do you conclude about the effect of class size on learning?
- b. Repeat part (a) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?
- c. Using children who are in either a regular-sized class or a regular-sized class with a teacher aide, estimate the regression model explaining students' combined aptitude scores as a function of the presence of a teacher aide, $TOTALSCORE = \gamma_1 + \gamma_2 AIDE + e$. Interpret the estimates. Based on this regression result, what do you conclude about the effect on learning of adding a teacher aide to the classroom?
- d. Repeat part (c) using dependent variables *READSCORE* and *MATHSCORE*. Do you observe any differences?

a.

$$\beta_1 = 916.4417$$

If the size is regular, the average total score is 916.4417

$$\beta_2 = 12.175$$

if the size is small: the average total score will increase 12.175

conclude: the small size is better than regular in student's learning

b.

read score:

$$\beta_1 = 432.665$$

$$\beta_2 = 6.92$$

math score:

$$\beta_1 = 483.78$$

$$\beta_2 = 5.25$$

if the size is small, the read score will increase more(6.92) then math score(5.25)

c.

$$\beta_1 = 916.4417$$

If the size is regular, the average total score is 916.4417

$$\beta_2 = 4.31$$

if the size is regular + aide: the average total score will increase 4.31

conclude: a TA will help students learning

d.

read score:

$$\beta_1 = 432.665$$

$$\beta_2 = 2.87$$

math score:

$$\beta_1 = 483.78$$

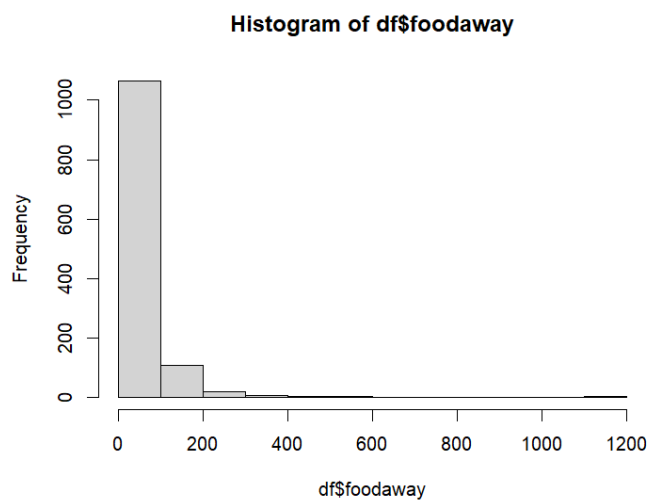
$$\beta_2 = 1.44$$

adding TA will help students more in reading(2.87) then math(1.44)

2.25 Consumer expenditure data from 2013 are contained in the file *cex5_small*. [Note: *cex5* is a larger version with more observations and variables.] Data are on three-person households consisting of a husband and wife, plus one other member, with incomes between \$1000 per month to \$20,000 per month. *FOODAWAY* is past quarter's food away from home expenditure per month per person, in dollars, and *INCOME* is household monthly income during past year, in \$100 units.

- a. Construct a histogram of *FOODAWAY* and its summary statistics. What are the mean and median values? What are the 25th and 75th percentiles?
- b. What are the mean and median values of *FOODAWAY* for households including a member with an advanced degree? With a college degree member? With no advanced or college degree member?
- c. Construct a histogram of $\ln(\text{FOODAWAY})$ and its summary statistics. Explain why *FOODAWAY* and $\ln(\text{FOODAWAY})$ have different numbers of observations.
- d. Estimate the linear regression $\ln(\text{FOODAWAY}) = \beta_1 + \beta_2 \text{INCOME} + e$. Interpret the estimated slope.
- e. Plot $\ln(\text{FOODAWAY})$ against *INCOME*, and include the fitted line from part (d).
- f. Calculate the least squares residuals from the estimation in part (d). Plot them vs. *INCOME*. Do you find any unusual patterns, or do they seem completely random?

a.



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	12.04	32.55	49.27	67.50	1179.00

mean = 49.27, median = 32.55, 25th = 12.04, 75th = 67.5

b.

advanced:

mean = 73.15

median = 48.15

college:

mean = 48.6

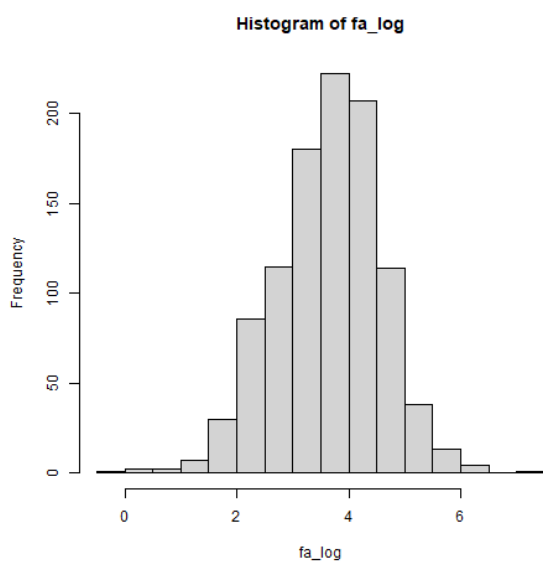
median = 36.11

No adv, coll

mean = 39.01

median = 26.02

c.



Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
-Inf	2.488	3.483	-Inf	4.212	7.072

因為 Foodaway 最小值是 0，取 log 會變成 -inf

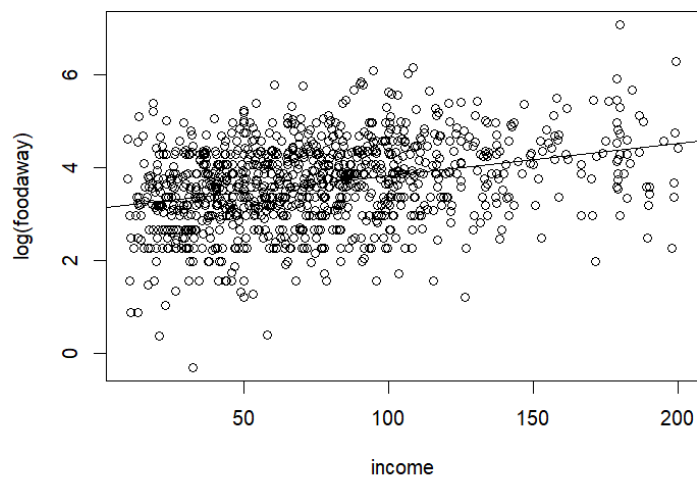
d.

$$\beta_1 = 3.13$$

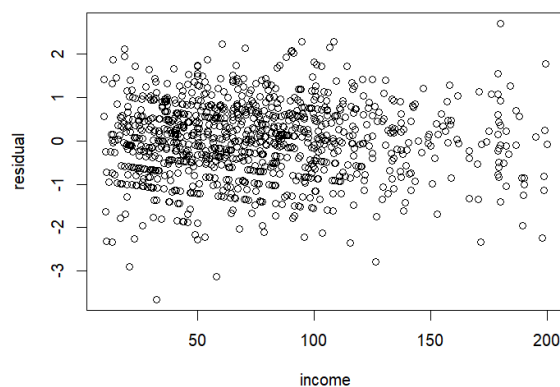
$$\beta_2 = 0.007$$

income increase a unit, foodaway will increase 0.7% in average.

e.



f.



The residuals look like a completely random